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Clinical challenges

Occam versus Hickam

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Case Report

A 23-year-old man was referred for new onset diplopia after an injury sustained 5 months prior. The patient had suffered a left orbital floor fracture confirmed by CT imaging after being punched in the face, He was also diagnosed with a concussion and diplopia; the diplopia resolved completely after one week. The orbital floor fracture was deemed too small to warrant surgical repair.

The patient had been at his baseline state of good health until 5 weeks prior to presentation, when he began to experience headaches and fatigue. He presented to an outside Emergency Room, where he underwent SARS-CoV-2 testing and a CT of the brain; both were unremarkable. One-and-a-half weeks prior to presentation, he developed diplopia and was referred for further evaluation. He described the diplopia as horizontal, with resolution when he covered either eye and no change with secondary gazes.

What are your thoughts on this chief complaint and timeline for referral?

Comments by Jurij R. Bilyk, MD:

This patient's presentation is unusual for an orbital floor fracture. The vast majority of diplopia from an orbital floor fracture occurs immediately after the injury. On occasion, patients may not notice diplopia for several days following the orbital injury because the affected eyelids are swollen shut; as the swelling dissipates, binocular diplopia may manifest.

Diplopia may certainly occur after orbital fracture repair, usually immediately following surgery. Rare cases of late-onset diplopia after fracture repair have been reported from hematoma, seroma, or abscess formation around the implant, but this patient never had orbital surgery. Large orbital fractures may cause slowly progressive enophthalmos and hypoglobus that mimics maxillary atelectasis ("silent sinus syndrome").[11] A small subset of such patients may complain of diplopia in extremes of gaze. Patients with midfacial or orbital fracture affecting the paranasal sinuses may also present with eventual sinusitis, which can spread into the orbit; however, this patient has no complaints of nasal congestion, midfacial pain, or eyelid edema/erythema. Furthermore, one would expect complaints of *vertical* diplopia from an orbital floor fracture; *horizontal* diplopia is usually associated with fracture of the lamina papyracea.

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Case report (continued)

The patient had no past ocular history other than the orbital floor fracture. His medical history was limited to asthma for which he used a beclomethasone inhaler. He had no family history of neurologic or ophthalmic conditions. His review of systems at the time of presentation was significant only for headaches and fatigue; he denied paresthesias, transient visual loss, memory or mentation problems, or any other neurologic symptoms. His mother, who accompanied the patient, had not noticed any recent mental status changes.

On examination, his uncorrected Snellen visual acuity was 20/20 in each eye, and he had equal and reactive pupils with no relative afferent pupillary defect, normal intraocular pressures, and full confrontation visual fields. Ishihara color plates were full and brisk bilaterally. On extraocular motility exam, the vertical ductions were normal; however, he was only able to abduct each eye 80% with a relatively commitant esotropia of 25 prism diopters and a right hypertropia of 5 prism diopters. There was no globe dystopia or asymmetry. The patient's anterior segments were unremarkable atthe slit lamp. There was no evidence of ptosis, eyelid retraction, lid lag, or lagophthalmos.

Given these exam findings, do you think the patient's diplopia is related to his orbital floor fracture?

Comments **by Dr. Bilyk (Continued):**

The patient had unilateral trauma to one orbital floor, but he has bilateral abduction deficits. The clinical findings do not match the previous history, with one exception: the patient had been diagnosed previously with a head injury and concussion. The patient's diplopia and external ophthalmoplegia are not related to the orbital floor fracture. At this juncture, they are also probably not related to his traumatic brain injury (TBI), with the rare exception of an intracranial bleed from a late posttraumatic vascular abnormality.

At this point, what does your differential diagnosis include?

Comments by Dr. Bilyk (Continued):

Unilateral or bilateral abduction deficits have a long differential. One common mistake is for the clinician to assume bilateral abduction deficits are equivalent to bilateral abducens nerve (CN-VI) palsies. CN-VI palsies represent only a subset of all abduction deficits. In such cases, it is prudent to step back and consider several main etiologies: restrictive, paralytic, congenital, neurogenic, and neuromuscular. In cases of trauma, restrictive strabismus from entrapment and/or scarring of the extraocular muscle (EOM) or surrounding fat is certainly of concern. Less frequently, EOM contusion or hematoma may cause acute restriction, and some of these cases will progress to intramuscular fibrosis. A second, common cause of a unilateral or bilateral abduction deficit is enlargement of the medial rectus muscle(s) from thyroid eye disease (TED). This patient has no other periocular stigmata of TED, but thyroid function and thyroid autoantibody testing can be considered. Congenital abnormalities such as Duane syndrome may manifest as an abduction deficit, but do not present with an acquired diplopia. Furthermore, the patient's mother would likely have mentioned this diagnosis, which is typically made in children. Neurogenic causes would include mitochondrial disease (chronic progressive external ophthalmoplegia, including Kearns-Sayre syndrome); however, patients with mitochondrial disease typically also have some degree of ptosis and more diffuse external ophthalmoplegia; of note, despite the EOM restriction, diplopia is not a typical complaint. Myasthenia gravis should always be considered in any patient with a strictly external ophthalmoplegia (e.g., without evidence of pupillary abnormalities – internal ophthalmoplegia).

Paralytic cases must also be considered. There is a broad differential for sixth nerve palsy/paresis, including autoimmune inflammatory conditions, demyelinating disease (multiple sclerosis), pachymeningitis, Tolosa Hunt syndrome, increased intracranial pressure, sequelae of intracranial trauma, vascular disease (e.g., vasculitis, brainstem infarct), infection (including a post-viral syndrome, especially in children), and neoplastic/paraneoplastic conditions.

Given this patient's history of trauma, are there any particular diagnoses you would be concerned about?

Comments by Dr. Bilyk (Continued)

An intracranial epidural hemorrhage is highly unlikely five months after initial injury. A subarachnoid hemmorrhage from a leaking posttraumatic vascular anomaly (such as an aneurysm) is but admittedly rare in relatively mild TBI. A subdural hemorrhage may occasionally slowly expand after trauma and may be exacerbated by an underlying coagulopathy or the use of systemic anticoagulants. A carotid-cavernous fistula (CCF) must also be considered, but would typically present with eyelid, conjunctival, and orbital findings; that said, some CCFs drain posteriorly into the deep cortical venous plexuses and present with a dearth of periocular abnormalities.

What diagnostic steps might you consider performing at this point?

Comments by Dr. Bilyk (Continued)

The first diagnostic step I would perform at this point would be to examine the posterior pole with a direct or indirect ophthalmoscope to assess for disc edema. This is a critical and accessible step that should be performed in every patient presenting with an abduction deficit to check for possible increased intracranial pressure.

There are also several other diagnostic steps which could help distinguish restrictive from paralytic strabismus. Forced ductions and force generations are easily performed with viscous lidocaine and toothed forceps but have limited value in patients with periocular edema and ecchymosis immediately after orbital trauma.

Other diagnostic steps that are less invasive include assessing speed of saccades, intraocular pressure measurement in different gazes, and the measurement of primary and secondary deviations.

Case report (continued)

On dilated fundus exam, the patient had circumferential disc edema in both eyes. The retina and macula were flat and without lesions in both eyes. His body mass index was normal.

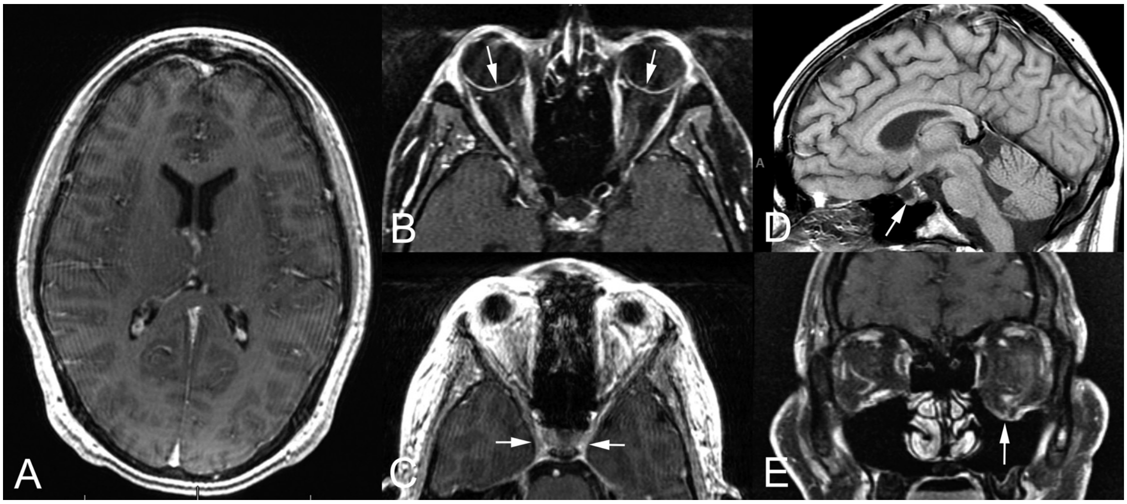

Fig. 1 – MRI, T1-weighted, postcontrast images. (A) Axial brain. No mass, edema, central venous sinus thrombosis, or other abnormalities is noted. (B) Axial orbits. There is no optic nerve enhancement or other abnormality. Mild flattening of the posterior sclera and disc edema are present (arrows). (C) Axial cavernous sinuses (arrows) show no abnormality or asymmetry. (D) Sagittal brain. Mild flattening of the pituitary gland and a partially empty sella turcica are present (arrow). (E) Coronal orbits. Evidence of a left orbital floor fracture with moderate soft tissue herniation is present (arrow).

What would your next steps be?

Comments by Dr. Bilyk (Continued)

Bilateral disc edema has a variety of causes, but the most urgently concerning is elevated intracranial pressure manifesting as papilledema. This patient's findings are a neurologic emergency. He requires immediate referral to an emergency room with same-day imaging. Work-up would include urgent magnetic resonance imaging (MRI) and MR venography (MRV) of the brain. It's important to include MRV with the MRI in all cases of suspected increased intracranial pressure to rule out central venous sinus thrombosis. If MRI/MRV are not readily available, a CT of the head should be performed to rule-out intracranial hemorrhage, midline shift, and impending brainstem herniation from mass effect. Neuro-ophthalmology should also be consulted – the patient's findings are not related to the orbit.

Case report (continued)

Same-day imaging was obtained, including an MRI brain/orbit with contrast and an MRV (Fig. 1). The MRI of the brain showed no acute intracranial abnormalities with a mild flattening of the pituitary gland and a partially empty sella. The MRI of the orbits demonstrated mild flattening of the globes and left orbital floor fracture with chronic changes and no significant orbital soft tissue herniation. No other abnormalities were noted. The MRV showed no thromboses, with hypoplasia and stenosis of the sigmoid and transverse sinuses.

Given these imaging results, what would your next steps be?

Comments by Adam DeBusk, DO

Some of these findings, such as a partially empty sella and hypoplasia of central venous sinuses can be seen in idiopathic intracranial hypertension (IIH) or be caused by elevated intracranial pressure. This patient has signs of elevated intracranial pressure with none of the typical risk factors for IIH: female gender, obesity, or use of certain systemic medications, but IIH is certainly still in the differential diagnosis. He has no mass lesion or venous sinus thrombosis on imaging. The next step is lumbar puncture to rule out intracranial inflammation, infection, or a neoplastic process of the meninges causing elevated intracranial pressure and to obtain an opening pressure.

Table 1 – A Biofire assay tests for fourteen common bacterial, fungal, and viral pathogens associated with central nervous system infections.

Cerebrospinal Fluid Results
Red blood cells: 1,991 (Tube 1), 226 (Tube 4)
White blood cells: 104 (Tube 1), 106 (Tube 4)
Glucose: 40
Protein: 124
Differential:
Neutrophils: 3%
Lymphocytes: 90%
Macrophages: 7%
Biofire: negative

Case report (continued)

A lumbar puncture was performed and the patient had a mildly elevated opening pressure of 25 cm H_2O; the tap was traumatic. Given the patient's increased intracranial pressure without any acute abnormalities, mass lesions, or thromboses on imaging, the patient was discharged home on acetazolamide 500 mg twice daily. The following day, the results of his cerebrospinal fluid (CSF) studies were reported (Table 1).

How would you interpret these CSF results?

Comments by Dr. DeBusk (Continued):

There is marked elevation in white blood cells, higher than would be expected if it were solely caused by contamination from the traumatic tap. The elevation is predominantly lymphocytic and is accompanied by a high protein level. This lymphocytic pleocytosis is less concerning for bacterial meningitis, in which I would expect the white blood cell count and neutrophils to be higher. Based on these CSF results, the differential diagnosis includes fungal disease, aseptic meningitis, and atypical meningitis such as syphilis, Lyme disease, and sarcoid. A central nervous system lymphoproliferative process is also possible. The lumbar puncture findings are not consistent with IIH.

Case report (continued)

Because of concern for meningitis, the patient was immediately admitted to Neurology. Upon admission, the patient was started on empiric intravenous antibiotics (ceftriaxone and vancomycin) and continued on acetazolamide. The Infectious Disease team was consulted, and a chest X-ray and additional infectious serologies were sent, all of which were unremarkable, including HIV, syphilis, and QuantiFERON gold. On the third day of the patient's admission, his CSF cultures were negative; antibiotics were discontinued. His headache improved, the diplopia remained stable off antibiotics for several days, and he was ultimately discharged home on acetazolamide. Several days after discharge, an elevated CSF Lyme antibody (4.02, normal <0.99) and increased IgG production (IgG index: 0.94, normal <0.66) were reported.

How do you interpret these results and what is your next step? Are intravenous antibiotics indicated in suspected Lyme meningitis?

Comments by Dr. DeBusk (Continued):

There is ongoing research evaluating oral versus intravenous antibiotics. The Quality Standards Subcommittee of the American Academy of Neurology published a review of 37 articles on antibiotic therapy for nervous system Lyme disease.[5] They noted that most studies used parenteral antibiotic regimens for neuroborreliosis; however, several European studies have demonstrated equivalent efficacy of oral doxycycline compared to intravenous ceftriaxone for adults with early disseminated disease, including meningitis.[5] There is variability in recommended duration of treatment and little data supporting prolonged antibiotic therapy, although this may be used in practice.[5] In general, Lyme meningitis responds well to antibiotic therapy.

Case report (concluded)

The Infectious Disease team reached out to the patient, who reported his headaches and his diplopia had significantly improved. Because of the possibility of Lyme meningitis, he was started on a 28-day course of oral doxycycline 100 mg twice daily. A serum Lyme Western Blot for *Borrelia burgdorferi* was obtained that was positive for both IgG and IgM.

The patient was examined 3 days after his hospital discharge and one day after starting doxycycline. At this time, he had persistent abduction deficits and optic nerve head edema. 24-2 Humphrey static fields showed enlarged blind spots in both eyes and no other abnormalities. He was continued on acetazolamide and doxycycline. Over the following 3 months, the patient's diplopia and headache completely resolved, he completed his oral doxycycline course, and his disc edema improved on acetazolamide (Fig. 2). He has remained asymptomatic for over 1 year.

Discussion

Meningitis is the most common manifestation of Lyme central nervous system involvement.[4] Lyme disease is caused by *Borrelia burgdorferi*, a spirochete transmitted via *Ixodes* ticks.[4] Similar to syphilis, which is also caused by a spirochete, Lyme disease is often described as having 3 stages: early localized disease (erythema migrans), early disseminated disease (which includes meningitis), and late disseminated disease.[1,4]

Diagnosing Lyme meningitis can be challenging, particularly in endemic areas where there are high rates of Lyme antibody seropositivity. Diagnosis is typically made by measuring patients' immune responses to the disease rather than direct measurement of the pathogen itself because of difficulty in culturing the organism.[6] Confirming Lyme meningitis requires demonstration of intrathecal production of Lyme antibodies, which is difficult in seropositive patients because serum IgG may enter the CSF. To account for this, diagnosis should involve simultaneous measurement of IgG titers from both the serum and CSF with subsequent dilution of samples until the total IgG is the same in both serum and CSF. A Lyme antibody titer that is higher in the CSF compared to serum indicates intrathecal production of antibodies.[6] For this patient, no serum antibody titer was drawn at the time of CSF collection. Therefore, although his elevated CSF IgG and Lyme antibody levels are highly suggestive of Lyme meningitis, they are not concretely diagnostic, and aseptic meningitis remains a possibility, albeit an unlikely one.

This patient's diagnosis was further complicated by his atypical presentation for Lyme meningitis. Fifteen percent of patients with Lyme disease develop meningitis, and the typical clinical presentation includes meningeal signs such as fevers, neck stiffness, photophobia, nausea/vomiting, which this patient lacked; rather, his findings mimicked those of IIH.[1] The patient's clinical presentation was also complicated by his previous blowout fracture, which was deemed to be the etiology of his diplopia by the referring physician.

There are a handful of case reports, mostly in children, describing patients with Lyme meningitis who presented with intracranial hypertension.[2,3,7–10] Importantly, the reports emphasize that patients often do not endorse a known history of a tick bite and that central nervous system involvement can occur without any reports of preceding cutaneous signs or flu-like symptoms. Overall, Lyme meningitis was found to have a good prognosis.

The use of oral antibiotic therapy for neuroborreliosis is still controversial, but a review of the literature suggests that oral therapy is a viable option for this condition.[5]

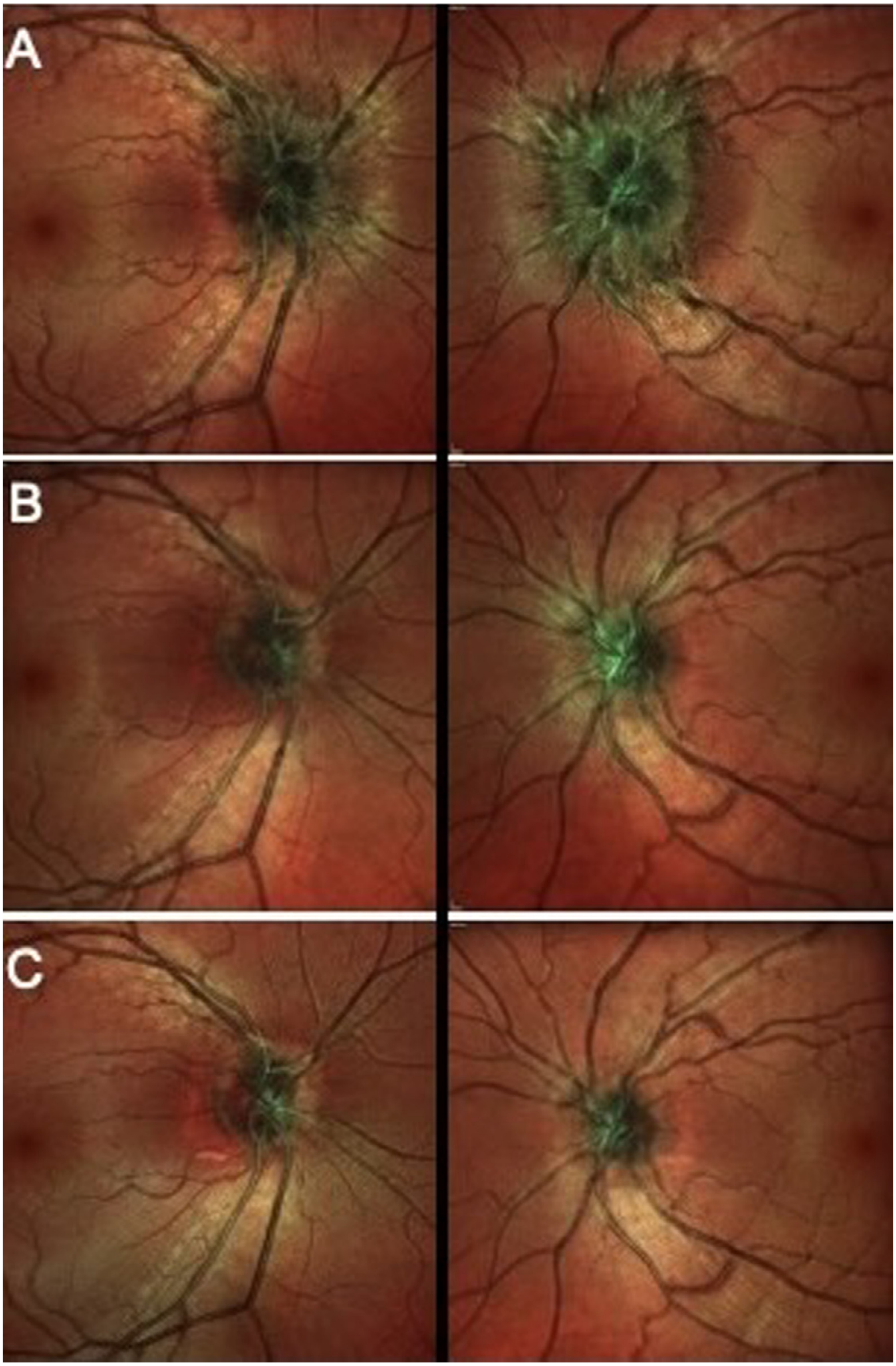

Fig. 2 – Multicolored optical coherence tomography of the patient's optic nerves. (A) Initial presentation. (B) Six weeks after presentation. (C) Three months after presentation. Note the resolution of the bilateral disc edema over time.

Conflicts of Interest

All the authors report no relevant conflicts of interest.

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ABSTRACT

A 23-year-old man presented with new onset horizontal diplopia 5 months after a left orbital floor fracture. Examination revealed bilateral abduction deficits and disc swelling. Urgent MRI and MRI showed no significant abnormalities in the CNS. Lumbar puncture revealed a minimally elevated opening pressure and significant leukocytosis. Additional CSF testing revealed probable Lyme meningitis. The patient responded to a course of oral doxycycline, with rapid resolution of his diplopia, abduction deficits, and disc edema.